

Rajalakshmi Engineering College

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Department: AI & ML - Section 4
Batch: 2028
Degree: B.E - AI & ML

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Rosh is intrigued by numerical patterns. Today, she stumbled upon a puzzle while working with arrays. She wants to compute the sum of the third-largest and second-smallest elements from a list of integers. She seeks your help to implement a program that solves this for her efficiently.

Input Format

The first line of input is an integer N, representing the size of the array.

The second line of input consists of N space-separated integers, representing the elements of the array.

Output Format

The output displays a single integer representing the sum of the third-largest and second-smallest elements in the array.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 10

10 20 30 40 50 60 70 80 90 100

Output: 100

Answer

```
// You are using Java
import java.util.Scanner;
import java.util.Arrays;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        int N = sc.nextInt(); // Size of the array
        int[] arr = new int[N];
```

```
        // Read array elements
        for (int i = 0; i < N; i++) {
            arr[i] = sc.nextInt();
        }
```

```
        // Sort the array
        Arrays.sort(arr);
```

```
        // Second-smallest is at index 1
        int secondSmallest = arr[1];
```

```
        // Third-largest is at index N - 3
        int thirdLargest = arr[N - 3];
```

```
        // Output the sum
        System.out.print(secondSmallest + thirdLargest);
```

```
        sc.close();
    }
```

}

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Monica is interested in finding a treasure but the key to opening is to get the sum of the main diagonal elements and secondary diagonal elements.

Write a program to help Monica find the diagonal sum of a square 2D array.

Note: The main diagonal of the array consists of the elements traversing from the top-left corner to the bottom-right corner. The secondary diagonal includes elements from the top-right corner to the bottom-left corner.

Input Format

The first line of input consists of an integer N, representing the number of rows and columns.

The following N lines consist of N space-separated integers, representing the 2D array elements.

Output Format

The first line of output prints "Sum of the main diagonal: " followed by an integer, representing the sum of the main diagonal.

The second line prints "Sum of the secondary diagonal: " followed by an integer, representing the sum of the secondary diagonal.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

1 2 3

4 5 6

7 8 9

Output: Sum of the main diagonal: 15

Sum of the secondary diagonal: 15

Answer

// You are using Java

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        int N = sc.nextInt(); // Size of the square matrix  
        int[][] matrix = new int[N][N];
```

```
        // Read the matrix elements  
        for (int i = 0; i < N; i++) {  
            for (int j = 0; j < N; j++) {  
                matrix[i][j] = sc.nextInt();  
            }  
        }
```

```
        int mainDiagonalSum = 0;
```

```
int secondaryDiagonalSum = 0;

for (int i = 0; i < N; i++) {
    mainDiagonalSum += matrix[i][i]; // Top-left to bottom-right
    secondaryDiagonalSum += matrix[i][N - 1 - i]; // Top-right to bottom-left
}

// Output as per formatting requirements
System.out.println("Sum of the main diagonal: " + mainDiagonalSum);
System.out.print("Sum of the secondary diagonal: " +
secondaryDiagonalSum);

    sc.close();
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

You are developing a warehouse management system for a shipping company. The system uses an integer array to represent the weights of packages in a specific order. To verify that the weight capacity is not exceeded, the program needs to calculate the sum of the weights of the first and last packages in the list.

Task:

Write a code to calculate the sum of the weights of the first and last packages in the list. The program should take an integer array as input and return the total weight of the first and last packages.

Input Format

The first line of the input is an integer N representing the size of the array.

The second line of the input is N space-separated integer values.

Output Format

The output is displayed in the following format:

"Sum of the first and last elements: <<Sum>>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

Output: Sum of the first and last elements: 60

Answer

// You are using Java

```
import java .util.Scanner;
```

```
public class Main{
```

```
    public static void main(String[]arg){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int N=sc.nextInt();
```

```
        int[] weight = new int [N];
```

```
        for (int i=0;i<N;i++){
```

```
            weight[i]=sc.nextInt();
```

```
        }
```

```
        int sum = weight[0]+weight[N-1];
```

```
        System.out.print("Sum of the first and last elements: "+sum);
```

```
    }
```

```
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sesha is developing a weather monitoring system for a region with multiple weather stations. Each weather station collects temperature data hourly and stores it in a 2D array.

Write a program that can add the temperature data from two different weather stations to create a combined temperature record for the region.

Input Format

The first line of input consists of two space-separated integers N and M, representing the number of rows and columns of the matrices, respectively.

The next N lines consist of M space-separated integers, representing the values of the first matrix.

The following N lines consist of M space-separated integers, representing the values of the second matrix.

Output Format

The output prints the addition of the two matrices in N rows and M columns, representing the combined temperature record.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3 3

1 2 3

4 5 6

7 8 9

1 1 1

2 2 2

3 3 3

Output: 2 3 4

6 7 8

10 11 12

Answer

```
// You are using Java
import java.util.Scanner;
```

```
public class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read dimensions of the matrix
        int N = sc.nextInt(); // Number of rows
        int M = sc.nextInt(); // Number of columns

        int[][] matrix1 = new int[N][M];
        int[][] matrix2 = new int[N][M];

        // Read first matrix
        for (int i = 0; i < N; i++) {
            for (int j = 0; j < M; j++) {
```

```
        matrix1[i][j] = sc.nextInt();
    }
}

// Read second matrix
for (int i = 0; i < N; i++) {
    for (int j = 0; j < M; j++) {
        matrix2[i][j] = sc.nextInt();
    }
}

// Add matrices and print result
for (int i = 0; i < N; i++) {
    for (int j = 0; j < M; j++) {
        System.out.print((matrix1[i][j] + matrix2[i][j]) + " ");
    }
    System.out.println();
}

sc.close();
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q5

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sharon is creating a program that finds the first repeated element in an integer array. The program should efficiently identify the first element that appears more than once in the given array. If no such element is found, it should appropriately display a message.

Help Sharon to complete the program.

Input Format

The first line of input consists of an integer n , representing the number of elements in the array.

The second line consists of n space-separated integers, representing the array elements.

Output Format

If a repeated element is found, print the first element that appears more than once.

If no repeated element is found, print "No repeated element found in the array".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 8

12 21 13 14 21 36 47 21

Output: 21

Answer

// You are using Java

import java.util.Scanner;

import java.util.HashSet;

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();  
        int[] arr = new int[n];
```

```
        for(int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }
```

```
        HashSet<Integer> seen = new HashSet<>();  
        int repeatedElement = -1;
```

```
        for(int num : arr) {  
            if(seen.contains(num)) {  
                repeatedElement = num;  
                break;  
            } else {  
                seen.add(num);  
            }
```

```
    }  
  }  
  
  if(repeatedElement == -1) {  
    System.out.print("No repeated element found in the array");  
  } else {  
    System.out.print(repeatedElement);  
  }  
}  
}
```

Status : Correct

Marks : 10/10